Q1 JEE Main 2020 - 7 Jan (Morning)

Let α be a root of equation $x^2+x+1=0$ and the matrix $A=\frac{1}{\sqrt{3}}\begin{bmatrix}1 & 1 & 1\\ 1 & \alpha & \alpha^2\\ 1 & \alpha^2 & \alpha^4\end{bmatrix}$, then the matrix A^{31} is equal to :

(A) A
(B) A^2
(C) A^3
(D) A^4

Q2 JEE Main 2020 - 7 Jan (Morning)

If the system of linear equations
$2x+2ay+az=0$
$2x+3by+bz=0$
$2x+4cy+cz=0$,
where $a,\ b,\ c\in R$ are non-zero and distinct ; has a non-zero solution, then

(A) $a+b+c=0$
(B) $a,\ b,\ c$ are in A.P.
(C) $\frac{1}{a},\ \frac{1}{b},\ \frac{1}{c}$ are in A.P.
(D) $a,\ b, c$ in G.P.

Q3 JEE Main 2020 - 7 Jan (Evening)

Let $A=[a_{ij}]$ and $B=[b_{ij}]$ be two 3×3 real matrices such that $b_{ij}=(3)^{(i+j-2)}a_{ji}$, where $i,\ j=1,\ 2,\ 3$. If the determinant of B is 81, then the determinant of A is:

(A) $\frac{1}{9}$
(B) 3
(C) $\frac{1}{81}$
(D) $\frac{1}{27}$

Q4 JEE Main 2020 - 7 Jan (Evening)

If the system of linear equations,
$x+y+z=6$
$x+2y+3z=10$
$3x+2y+\lambda z=\mu$

has more than two solutions, then $\mu - \lambda^2$ is equal to . . . . .

Q5 JEE Main 2020 - 8 Jan (Morning)

For which of the following ordered pairs $(\mu,\ \delta)$, the system of linear equations

$x + 2y + 3z = 1$

$3x + 4y + 5z = \mu$

$4x + 4y + 4z = \delta$

is inconsistent?

(A) $(4,\ 6)$
(B) $(3,\ 4)$
(C) $(4,\ 3)$
(D) $(1,\ 0)$

Q6 JEE Main 2020 - 8 Jan (Morning)

The number of all 3×3 matrices A, with entries from the set $\{-1,\ 0,\ 1\}$ such that the sum of the diagonal elements of AA^T is 3, is . . . . .

Q7 JEE Main 2020 - 8 Jan (Evening)

If $A = \begin{pmatrix} 2 & 2 \\ 9 & 4 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, then $10A^{-1}$ is equal to :

(A) $4I - A$
(B) $6I - A$
(C) $A - 4I$
(D) $A - 6I$

Q8 JEE Main 2020 - 8 Jan (Evening)

The system of linear equations

$\lambda x + 2y + 2z = 5$

$2\lambda x + 3y + 5z = 8$

$4x + \lambda y + 6z = 10 \text{ has}$

(A) Infinite solutions when $\lambda = 8$
(B) Infinite solutions when $\lambda = 2$
(C) no solutions when $\lambda = 8$
(D) no solutions when $\lambda = 2$

Q9 JEE Main 2020 - 9 Jan (Morning)

If $A=\begin{bmatrix}1 & 1 & 2\\ 1 & 3 & 4\\ 1 & -1 & 3\end{bmatrix}$, $B=adj\,(A)$ and $C=3A$ then $\frac{|adj\,B|}{|c|}$ is

(A) 8
(B) 4
(C) 2
(D) 16

Q10 JEE Main 2020 - 9 Jan (Evening)

Let $a-2b+c=1$,

If $f(x)=\begin{vmatrix}x+a & x+2 & x+1\\ x+b & x+3 & x+2\\ x+c & x+4 & x+3\end{vmatrix}$, then:

(A) $f(-50)=-1$
(B) $f(50)=1$
(C) $f(50)=-501$
(D) $f(-50)=501$

Q11 JEE Main 2020 - 9 Jan (Evening)

The following system of linear equations

$7x+6y-2z=0$
$3z+4y+2z=0$
$x-2y-6z=0,\ \text{has}$

(A) infinitely many solution, $(x,\ y,\ z)$ satisfying $y=2\ z$
(B) infinitely many solution, $(x,\ y,\ z)$ satisfying $x=2\ z$
(C) Only the trivial solution.
(D) no solution.

Q12 JEE Main 2020 - 2 Sep (Morning)

Let S be the set of all $\lambda\in R$ for which the system of linear equations

$2x-y+2z=2$
$x-2y+\lambda z=-4$
$x+\lambda y+z=4$

has no solution. Then the set S

(A) is a singleton.
(B) contains more than two elements.
(C) is an empty set.
(D) contains exactly two elements.

Q13 JEE Main 2020 - 2 Sep (Morning)

Let A be a 2×2 real matrix with entries from \{0,1\} and $|A| \neq 0$. Consider the following two statements :
(P) If $A \neq I_2$, then $|A| = -1$
(Q) If $|A| = 1$, then $\operatorname{tr}(A) = 2$,
where I_2 denotes 2×2 identity matrix and $\operatorname{tr}(A)$ denotes the sum of the diagonal entries of A.
Then :

(A) (P) is true and (Q) is false
(B) Both (P) and (Q) are false
(C) Both (P) and (Q) are true
(D) (P) is false and (Q) is true

Q14 JEE Main 2020 - 2 Sep (Evening)

Let $a, b, c \in R$ be all non-zero and satisfy $a^3 + b^3 + c^3 = 2$. If the matrix $A = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$ satisfies $A^\top A = 1$, then a value of abc can be :

(A) 3
(B) $\frac{1}{3}$
(C) $-\frac{1}{3}$
(D) $\frac{2}{3}$

Q15 JEE Main 2020 - 2 Sep (Evening)

Let $A = \{X = (x, y, z)^\top : PX = 0$ and $x^2 + y^2 + z^2 =$1\}, where $P = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 3 & -4 \\ 1 & 9 & -1 \end{bmatrix}$, then the set A :

(A) is an empty set.
(B) contains more than two elements.

(C) contains exactly two elements.
(D) is a singleton.

Q16 JEE Main 2020 - 3 Sep (Morning)

If $\Delta=\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix}=Ax^3+Bx^2+Cx+D$, then $B+C$ is equal to :

(A) 9
(B) -1
(C) 1
(D) -3

Q17 JEE Main 2020 - 3 Sep (Morning)

Let $A=\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$, $x \in R$ and $A^4=\left[a_{ij}\right]$. If $a_{11}=109$, then a_{22} is equal to

Q18 JEE Main 2020 - 3 Sep (Evening)

Let A be a 3×3 matrix such that $\operatorname{adj} A=\begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{bmatrix}$ and $B=\operatorname{adj}(adjA)$ If $|A|=\lambda$ and $\left|\left(B^{-1}\right)^{\top}\right|=\mu$, then the ordered pair,$(|\lambda|, \mu)$ is equal to

(A) (3,81)
(B) $\left(9, \frac{1}{9}\right)$
(C) $\left(3, \frac{1}{81}\right)$
(D) $\left(9, \frac{1}{81}\right)$

Q19 JEE Main 2020 - 3 Sep (Evening)

Let S be the set of all integer solutions, (x, y, z), of the system of equations
$x-2y+5z=0$
$-2x+4y+z=0$
$-7x+14y+9z=0$
such that $15 \leq x^2+y^2+z^2 \leq 150$. Then, the number of elements in the set S is equal to

Q20 JEE Main 2020 - 4 Sep (Morning)

If $A = \begin{bmatrix} \cos\theta & \text{i}\sin\theta \\ i\sin\theta & \cos\theta \end{bmatrix}$, $\left(\theta = \frac{\pi}{24}\right)$ and $A^5 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $i = \sqrt{-1}$, then which one of the following is not true?

(A) $a^2 - b^2 = \frac{1}{2}$
(B) $a^2 - c^2 = 1$
(C) $a^2 - d^2 = 0$
(D) $0 \leq a^2 + b^2 \leq 1$

Q21 JEE Main 2020 - 4 Sep (Morning)

If the system of equations
$x - 2y + 3z = 9$
$2x + y + z = b$
$x - 7y + az = 24$, has infinitely many solutions, then $a - b$ is equal to

Q22 JEE Main 2020 - 4 Sep (Evening)

If the system of equations
$x + y + z = 2$
$2x + 4y - z = 6$
$3x + 2y + \lambda z = \mu$
has infinitely many solutions, then:

(A) $2\lambda - \mu = 5$
(B) $\lambda - 2\mu = -5$
(C) $\lambda + 2\mu = 14$
(D) $2\lambda + \mu = 14$

Q23 JEE Main 2020 - 4 Sep (Evening)

Suppose the vectors x_1, x_2 and x_3 are the solutions of the system of linear equations, $Ax = b$ when the vector b on the right side is equal to b_1, b_2 and b_3 respectively. If $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$ and $\mathbf{b}_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$, then the determinant of A is equal to:

(A) 4

(B) $\frac{1}{2}$
(C) 2
(D) $\frac{3}{2}$

Q24 JEE Main 2020 - 5 Sep (Morning)

Let $\lambda \in \mathbb{R}$. The system of linear equations
$2x_1 - 4x_2 + \lambda x_3 = 1$
$x_1 - 6x_2 + x_3 = 2$
$\lambda x_1 - 10x_2 + 4x_3 = 3$
is inconsistent for :

(A) exactly two values of λ.
(B) exactly one positive value of λ.
(C) every value of λ.
(D) exactly one negative value of λ.

Q25 JEE Main 2020 - 5 Sep (Morning)

If the minimum and the maximum values of the function $f : \left[\frac{\pi}{4}, \frac{\pi}{2}\right] \rightarrow R$, defined by

$f(\theta) = \begin{vmatrix} -\sin^2\theta & -1-\sin^2\theta & 1 \\ -\cos^2\theta & -1-\cos^2\theta & 1 \\ 12 & 10 & -2 \end{vmatrix}$ are m and M respectively, then the ordered pair (m, M) is equal to :

(A) $(0, 2\sqrt{2})$
(B) $(0, 4)$
(C) $(-4, 4)$
(D) $(-4, 0)$

Q26 JEE Main 2020 - 5 Sep (Evening)

If $a + x = b + y = c + z + 1$, where a, b, c, x, y, z are non-zero distinct real numbers, then $\begin{vmatrix} x & a+y & x+a \\ y & b+y & y+b \\ z & c+y & z+c \end{vmatrix}$ is equal to :

(A) $y(b-a)$
(B) $y(a-b)$
(C) $y(a-c)$

(D) 0

Q27 JEE Main 2020 - 5 Sep (Evening)

If the system of linear equations
$x+y+3z=0$
$x+3y+k^2z=0$
$3x+y+3z=0$
has a non-zero solution (x, y, z) for some $k \in R$ then $x+\left(\frac{y}{z}\right)$ is equal to

(A) 9
(B) 3
(C) -9
(D) -3

Q28 JEE Main 2020 - 6 Sep (Morning)

Let m and M be respectively the minimum and maximum values of
$\left|\begin{array}{ccc}\cos^2 x & 1+\sin^2 x & \sin 2x \\ 1+\cos^2 x & \sin^2 x & \sin 2x \\ \cos^2 x & \sin^2 x & 1+\sin 2x\end{array}\right|$ Then the ordered pair (m, M) is equal to :

(A) (1,3)
(B) (-3,-1)
(C) (-4,-1)
(D) (-3,3)

Q29 JEE Main 2020 - 6 Sep (Morning)

The values of λ and μ for which the system of linear equations
$x+y+z=2$
$x+2y+3z=5$
$x+3y+\lambda z=\mu$
has infinitely many solutions are, respectively

(A) 5 and 7
(B) 6 and 8
(C) 4 and 9
(D) 5 and 8

Q30 JEE Main 2020 - 6 Sep (Evening)

Let $\theta=\frac{\pi}{5}$ and $A=\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$. If $B=A+A^4$ then det (B) :

(A) lies in (2,3)
(B) is zero.
(C) is one.
(D) lies in (1,2)

Q31 JEE Main 2020 - 6 Sep (Evening)

The sum of distinct values of λ for which the system of equations
$(\lambda-1)x+(3\lambda+1)y+2\lambda z=0$
$(\lambda-1)x+(4\lambda-2)y+(\lambda+3)z=0$
$2x+(3\lambda+1)y+3(\lambda-1)z=0$
has non-zero solutions, is

Answer Key

Q1 (C)	**Q2** (C)	**Q3** (A)	**Q4** (13)
Q5 (C)	**Q6** (672)	**Q7** (D)	**Q8** (D)
Q9 (A)	**Q10** (B)	**Q11** (B)	**Q12** (D)
Q13 (D)	**Q14** 2^*	**Q15** (C)	**Q16** (D)
Q17 (10)	**Q18** (C)	**Q19** (8)	**Q20** (A)
Q21 (5)	**Q22** (D)	**Q23** (C)	**Q24** (D)
Q25 (D)	**Q26** (B)	**Q27** (D)	**Q28** (B)
Q29 (D)	**Q30** (D)	**Q31** (3)	

Q1

$$A^2 = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow \quad A^4 = I$$

$$\Rightarrow A^{30} = A^{28} \times A^3 = A^3$$

Q2

For non-trivial solution

$$\begin{vmatrix} 2 & 2a & a \\ 2 & 3b & b \\ 2 & 4c & c \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4c & c \end{vmatrix} = 0$$

$$(3bc - 4bc) - (2ac - 4ac) + (2ab - 3ab) = 0$$

$$-bc + 2ac - ab = 0$$

$$ab + bc = 2ac$$

$a,\ b,\ c$ in H.P.

$$\Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c} \text{ in A.P.}$$

Q3

$$|B| = \begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix} = \begin{vmatrix} 3^0 a_{11} & 3^1 a_{12} & 3^2 a_{13} \\ 3^1 a_{21} & 3^2 a_{22} & 3^3 a_{23} \\ 3^2 a_{31} & 3^3 a_{32} & 3^4 a_{33} \end{vmatrix}$$

$$\Rightarrow \quad 81 = 3^3 \cdot 3 \cdot 3^2\,|A|$$

$$\Rightarrow \quad 3^4 = 3^6\,|A| \quad \Rightarrow \quad |A| = \frac{1}{9}$$

Q4

$x+y+z=6 \quad \ldots(1)$
$x+2y+3z=10 \quad \ldots(2)$
$3x+2y+\lambda z=\mu \quad \ldots(3)$
from (1) and (2)
if $z=0 \Rightarrow x+y=6$ and $x+2y=10$
$\Rightarrow y=4,\ x=2$
$(2,\ 4,\ 0)$
if $y=0 \Rightarrow x+z=6$ and $x+3z=10$
$\Rightarrow z=2$ and $x=4$
$(4,\ 0,\ 2)$
so $3x+2y+\lambda z=\mu$
must pass through $(2,\ 4,\ 0)$ and $(4,\ 0,\ 2)$
so $6+8=\mu \Rightarrow \mu=14$
so $12+2\lambda=\mu$
$12+2\lambda=14 \Rightarrow \lambda=1$
so $\mu-\lambda^2=14-1=13$

Q5

$$\text{Note } D=\begin{vmatrix}3&4&5\\1&2&3\\4&4&4\end{vmatrix}(R_3\to R_3-2R_1+3R_2)$$
$$=\begin{vmatrix}3&4&5\\1&2&3\\0&0&0\end{vmatrix}=0$$
Now let $P_3\equiv 4x+4y+4z-\delta=0$.
If the system has solutions it will have infinite solution,
so $P_3\equiv\alpha P_1+\beta P_2$
Hence $3\alpha+\beta=4$ & $4\alpha+2\beta=4$
$\Rightarrow \alpha=2$ & $\beta=-2$
So for infinite solution $2\mu-2=\delta$
$\Rightarrow$ for $2\mu\neq\delta+2$
system inconsistent

Q6

Let $A = [a_{ij}]_{3\times 3}$

$tr\left(AA^T\right) = 3$

$a_{11}^2 + a_{12}^2 + a_{13}^2 + a_{21}^2 + \ldots + a_{33}^2 = 3$

possible cases

$$\left.\begin{array}{rl} 0,0,0,0,0,0,1,1,1 & \to 1 \\ 0,0,0,0,0,0,-1,-1,-1 & \to 1 \\ 0,0,0,0,0,0,1,1-1 & \to 3 \\ 0,0,0,0,0,0,-1,1,-1 & \to 3 \end{array}\right\} {}^9C_6 \times 8 = 84 \times 8 = 672$$

Q7

Characteristics equation of matrix A is

$$\begin{vmatrix} 2-x & 2 \\ 9 & 4-x \end{vmatrix} = 0$$

$\Rightarrow \; x^2 - 6x - 10 = 0$

$\therefore \; A^2 - 6A - 10I = 0$

$\Rightarrow 10A^{-1} = A - 6I$

Q8

$$D = \begin{vmatrix} \lambda & 2 & 2 \\ 2\lambda & 3 & 5 \\ 4 & \lambda & 6 \end{vmatrix}$$

$D = (\lambda + 8)(2 - \lambda)$

for $\lambda = 2$

$$D_1 = \begin{vmatrix} 5 & 2 & 2 \\ 8 & 3 & 5 \\ 10 & 2 & 6 \end{vmatrix}$$

$= 5[18 - 10] - 2[48 - 50] + 2(16 - 30)$

$= 40 + 4 - 28 \neq 0$

No solutions for $\lambda = 2$

Q9

$$|A| = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{vmatrix} = ((9+4) - 1(3-4) + 2(-1-3))$$
$$= 13 + 1 - 8 = 6$$
$$|adj\ B| = |adj\ adj\ A| = |A|^{(n-1)^2} = |A|^4 = (36)^2$$
$$|C| = |BA| = 3^3 \times 6$$
$$\frac{|adj\ B|}{|C|} = \frac{36 \times 36}{3^3 \times 6} = 8$$

Q10

Apply $R_1 = R_1 + R_3 - 2R_2$

$$\Rightarrow f(x) = \begin{vmatrix} 1 & 0 & 0 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix} \Rightarrow f(x) = 1$$
$$\Rightarrow f(50) = 1$$

Q11

$$\begin{vmatrix} 7 & 6 & -2 \\ 3 & 4 & 2 \\ 1 & -2 & -6 \end{vmatrix}$$
$$= 7(-20) - 6(-20) - 2(-10)$$
$$= -140 + 120 + 20 = 0$$

so infinite non-trivial solution exist

now equation $(1) + 3$ equation (3)

$$10x - 20z = 0$$
$$x = 2z$$

Q12

$$\begin{vmatrix} 2 & -1 & 2 \\ 1 & -2 & \lambda \\ 1 & \lambda & 1 \end{vmatrix} = 0$$
$$\Rightarrow 2\lambda^2 - \lambda - 1 = 0$$

$\Rightarrow \lambda = 1$ or $-\frac{1}{2}$

When $\lambda = 1$

$2x-y+2z=2$ ..(1)
$x-2y+z=-4$ ..(2)
$x+y+z=4$ ..(3)
Adding (2) and (3), we get
$2x-y+2z=0$ (contradiction) hence no solution.
When $\lambda=-\frac{1}{2}$
$2x-y+2z=2$ ..(1)
$x-2y-\frac{1}{2}z=-4$ ..(2)
$x-\frac{1}{2}y+z=4$ ..(3)
(1) and (3) contradict each other, hence no solution.

Q13

Let $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $a, b, c, d \in \{0,1\}$
$\Rightarrow |A|=ad-bc$
$\therefore \quad ad=0$ or 1 and $bc=0$ or 1
so possible values of $|\mathrm{A}|$ are 1,0 or -1
(P) If $A \neq I_2$ then $|A|$ is either 0 or -1
(Q) If $|A|=1$ then $ad=1$ and $bc=0$
$\Rightarrow a=d=1 \Rightarrow \operatorname{Tr}(A)=2$

Q14

$\because AA^{\top}=1 \Rightarrow |A|=\pm 1$
$|A|=3abc-\left(a^3+b^3+c^3\right)=\pm 1$
$$\left\{\because a^3+b^3+c^3=2\right\}$$
$\Rightarrow 3abc=3$ or 1
$\Rightarrow \quad$ abc $=\frac{1}{3}$ or 1
$*$ For the data given in the question, there exist no real numbers a, b and c; because
$AA^{\top}=1 \Rightarrow a^2+b^2+c^2=1$ and $ab+bc+ca=0$
also $a^3+b^3+c^3=2$ leads to no solutions.

Q15

$\because \det(P)=0$
So the system has infinitely many solutions.
All solution lies on the line of intersection of

planes
$x+2y+z=0, -2x+3y-4z=0$ and $x+9y-z=0$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ -2 & 3 & -4 \end{vmatrix} = -11\hat{i}+2\hat{j}+7\hat{k}$$

So, $x=-11\lambda, y=2\lambda, z=7\lambda$
$\because x^2+y^2+z^2=1$
$\Rightarrow \lambda^2=\frac{1}{174} \Rightarrow \lambda=\pm\frac{1}{\sqrt{174}}$
Two values of λ gives two triplets of (x,y,z)

Q16

$$\Delta=\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix}$$

$$\Rightarrow \Delta=\begin{vmatrix} x-2 & x-1 & x-1 \\ 2x-3 & x-1 & x-1 \\ 3x-5 & 2x-3 & 5x-9 \end{vmatrix} \quad \begin{matrix} c_3 \rightarrow c_3-c_2 \\ c_2 \rightarrow c_2-c_1 \end{matrix}$$

$$\Rightarrow \Delta=\begin{vmatrix} x-2 & x-1 & x-1 \\ x-1 & 0 & 0 \\ 3x-5 & 2x-3 & 5x-9 \end{vmatrix} \quad R_2 \rightarrow R_2-R_1$$

$$\Rightarrow \quad \Delta=-(x-1)\left[\left(5x^2-14x+9\right)-\left(2x^2-5x+3\right)\right]$$

$$=-3x^3+12x^2-15x+6$$

$$So; B+C=-3$$

Q17

$$A^2=\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}=\begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}$$

$$A^4=\begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}\begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}$$

$$\begin{bmatrix} \left(x^2+1\right)^2+x^2 & x\left(x^2+1\right)+x \\ x\left(x^2+1\right)+x & x^2+1 \end{bmatrix}$$

Given $\left(x^2+1\right)^2+x^2=109$
Let $x^2+1=t$
$t^2+t-1=109$

$(t-10)(t+11)=0$
$t=10=x^2+1=a_{22}$

Q18

$|\text{ adj } A|=2(4)+1(1-2)+1(2)$
$|\text{adj } A|=9$
and $|\text{ adj } A|=|A|^{n-1}=9$
$\Rightarrow \lambda^2=9 \Rightarrow \lambda=\pm 3 \quad \{\because |A|=\lambda\}$
Now
$|\mathbf{B}|=|\mathbf{adj}(\mathbf{adj}\mathbf{A})|=|\mathbf{A}|^{(\mathbf{n}-1)^2}=\lambda^4=3^4=\mathbf{81}$
Now
$\mu=\left|\left(B^{-1}\right)^{\top}\right|=\left|\left(B^{\top}\right)^{-1}\right|=\left|\left(B^{\top}\right)\right|^{-1}=|B|^{-1}=\frac{1}{|B|}=\frac{1}{81}$
So $(|\lambda|,\mu)=\left(3,\frac{1}{81}\right)$

Q19

$x-2y+5z=0$ (i)
$-2x+4y+z=0$ (ii)
$-7x+14y+9z=0$ (iii)
From (i) and (ii); $z=0$ and $x=2y$
Let $x=2\alpha, y=\alpha, z=0$
Now
$15 \leq 4\alpha^2+\alpha^2 \leq 150$
$3 \leq \alpha^2 \leq 30$
$\alpha=\pm 2, \pm 3, \pm 4, \pm 5$
Hence 8 elements are there in set S.

Q20

$\because A=\begin{bmatrix}\cos\theta & i\sin\theta \\ i\sin\theta & \cos\theta\end{bmatrix}$
$\therefore \quad A^n=\begin{bmatrix}\cos n\theta & i\sin n\theta \\ i\sin n\theta & \cos n\theta\end{bmatrix}, n \in N$
$\therefore \quad A^5=\begin{bmatrix}\cos 5\theta & i\sin 5\theta \\ i\sin 5\theta & \cos 5\theta\end{bmatrix}=\begin{bmatrix}a & b \\ c & d\end{bmatrix}$
$\therefore \quad a=\cos 5\theta, b=i\sin 5\theta=c, d=\cos 5\theta$
$\therefore \quad a^2-b^2=\cos^2 5\theta+\sin^2 5\theta=1$

$a^2 - c^2 = \cos^2 5\theta + \sin^2 5\theta = 1$
$a^2 - d^2 = \cos^2 5\theta - \cos^2 5\theta = 1$
$a^2 + b^2 = \cos^2 5\theta - \sin^2 5\theta = \cos 10\theta = \cos \frac{10\pi}{24}$
and $0 < \cos \frac{5\pi}{12} < 1 \quad \Rightarrow 0 \le a^2 + b^2 \le 1$
$\therefore$ option (1) is not true

Q21

$\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$
[For infinite solutions]

$$\Delta = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 1 & 1 \\ 1 & -7 & a \end{vmatrix} = 0$$

$\Rightarrow (a+7) - 2(1-2a) + 3(-15) = 0$
$\Rightarrow a = 8$

$$\Delta_3 = \begin{vmatrix} 1 & -2 & 9 \\ 2 & 1 & b \\ 1 & -7 & 24 \end{vmatrix} = 0$$

$\Rightarrow \quad (24 + 7b) - 2(b - 48) + 9(-15) = 0$
$\Rightarrow \quad b = 3$
$\therefore a - b = 5$

Q22

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & -1 \\ 3 & 2 & \lambda \end{vmatrix} = 0$$

$\Rightarrow -15 + 6 + 2\lambda = 0$
$\Rightarrow \quad \lambda = \frac{9}{2}$
Substituting the value of λ in equations, we get
$x + y + z = 2$
$2x + 4y - z = 6$
$6x + 4y + 9z = 2\mu$
(1) $\times 8 - (2)$ gives, $6x + 4y + 9z = 10$
So for infinitely many solutions, $2\mu = 10$
$\Rightarrow \mu = 5$

Q23

$A\times=b$ has solutions x_1,x_2,x_3
$\therefore\quad x_1+y_1+z_1=1$
$2y_1+z_1=2$
$z_1=2$
Above system equation has solution
Here $|\mathbf{A}|=\begin{vmatrix}1&1&1\\0&2&1\\0&0&1\end{vmatrix}$
$=2$

Q24

$\because\begin{vmatrix}2&-4&\lambda\\1&-6&1\\\lambda&-10&4\end{vmatrix}=0\Rightarrow 3\lambda^2-7\lambda-12=0$
$\Rightarrow\quad\lambda=3$ or $-\frac{2}{3}$
Adding first two equations, we get
$3x_1-10x_2+(\lambda+1)x_3=3$
and the last equation is $\lambda x_1-10x_2+4x_3=3$
So, for $\lambda=3$ there will be infinitely many
solutions and for $\lambda=-\frac{2}{3}$ there will be no solution (i.e. equations will be inconsistent).

Q25

$f(\theta)=\begin{vmatrix}-\sin^2\theta&-1-\sin^2\theta&1\\-\cos^2\theta&-1-\cos^2\theta&1\\12&10&-2\end{vmatrix}$
$\Rightarrow f(\theta)=4\cos2\theta$ (after finding determinant)
$\Rightarrow f'(\theta)=-8\sin2\theta<0\quad\forall\theta\in\left[\frac{\pi}{4},\frac{\pi}{2}\right]$
So f is decreasing function
So $f(\theta)_{\min}=f\left(\frac{\pi}{2}\right)=4\times\cos\pi=-4=m$
$f(\theta)_{\max}=f\left(\frac{\pi}{4}\right)=4\cos\frac{\pi}{2}=0=M$
So $(\mathrm{m},\mathrm{M})=(-4,0)$

Q26

$$\begin{vmatrix} x & a+y & x+a \\ y & b+y & y+b \\ z & c+y & z+c \end{vmatrix} = \begin{vmatrix} x & a & x+a \\ y & b & y+b \\ z & c & z+c \end{vmatrix} + y\begin{vmatrix} x & 1 & x+a \\ y & 1 & y+b \\ z & 1 & z+c \end{vmatrix}$$

$$= 0 + y\begin{vmatrix} x & 1 & x+a \\ y-x & 0 & 0 \\ z-x & 0 & -1 \end{vmatrix} \quad \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix}$$

$= -y(x-y) = -y(b-a)$

$= y(a-b)$

Q27

Here $\begin{vmatrix} 1 & 1 & 3 \\ 1 & 3 & k^2 \\ 3 & 1 & 3 \end{vmatrix} = 0$

$1\left(9-k^2\right) - 1\left(3-3k^2\right) + 3(1-9) = 0$

$9 - k^2 - 3 + 3k^2 - 24 = 0$

$2k^2 = 18 \Rightarrow k^2 = 9, k = \pm 3$

So equations are

$x + y + 3z = 0$ (i)

$x + 3y + 9z = 0$ (ii)

$3x + y + 3z = 0$ (iii)

Now (i) - (ii)

$-2y - 6z = 0 \Rightarrow y = -3z \Rightarrow \frac{y}{z} = -3$ (iv)

Now,

(i) $-$ (iii)

$-2x = 0 \Rightarrow x = 0$

So $x + \frac{y}{z} = 0 - 3 = -3$

Q28

Let $f(x) = \begin{vmatrix} \cos^2 x & 1+\sin^2 x & \sin 2x \\ 1+\cos^2 x & \sin^2 x & \sin 2x \\ \cos^2 x & \sin^2 x & 1+\sin 2x \end{vmatrix}$

After expanding it we get

$f(x) = -2 - \sin 2x$

$f'(x) = -2\cos 2x = 0$

$\Rightarrow \cos 2x = 0$

$2x = \frac{\pi}{2}, \frac{3\pi}{2}$
$x = \frac{\pi}{4}, \frac{3\pi}{4}$
$f'(x) = 4\sin 2x$
So $f''\left(\frac{\pi}{4}\right) = 4 > 0$ (minima)
$f''\left(\frac{3\pi}{4}\right) = -4 < 0$ (maxima)
So $\mathrm{m} = \mathrm{f}_{\min} = \mathrm{f}\left(\frac{\pi}{4}\right) = -2 - 1 = -3$
and $\mathrm{M} = \mathrm{f}_{\max} = \mathrm{f}\left(\frac{3\pi}{4}\right) = -2 + 1 = -1$
So $(\mathrm{m}, \mathrm{M}) = (-3, -1)$

Q29

Here $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & \lambda \end{vmatrix} = 0$
$\Rightarrow \lambda = 5$
and also
$\Delta_1 = \begin{vmatrix} 2 & 1 & 1 \\ 5 & 2 & 3 \\ \mu & 3 & 5 \end{vmatrix} = 0$
$\Rightarrow \mu = 8$

Q30

$\because A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$
$\therefore \quad A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}, n \in N$
$\therefore \quad B = A + A^4 = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} + \begin{bmatrix} \cos 4\theta & \sin 4\theta \\ -\sin 4\theta & \cos 4\theta \end{bmatrix}$

$$B=\begin{bmatrix}\cos\frac{\pi}{5}+\cos\frac{4\pi}{5} & \sin\frac{\pi}{5}+\sin\frac{4\pi}{5}\\ -\sin\frac{\pi}{5}-\sin\frac{4\pi}{5} & \cos\frac{\pi}{5}+\cos\frac{4\pi}{5}\end{bmatrix}$$

$$\begin{aligned}\det(\mathbf{B}) &= 2\sin\left(\frac{\pi}{5}\right)\cdot\begin{vmatrix}0 & 1\\ -1 & 0\end{vmatrix}\\ &= \frac{\sqrt{10-2\sqrt{5}}}{2}\approx\frac{2.35}{2}\simeq 1.175\end{aligned}$$

$$\det\mathbf{B}\in(1,2)$$

Q31

Here $\begin{vmatrix}\lambda-1 & 3\lambda+1 & 2\lambda\\ \lambda-1 & 4\lambda-2 & \lambda+3\\ 2 & 3\lambda+1 & 3(\lambda-1)\end{vmatrix}=0$

On solving it we get

$6\lambda^3-36\lambda^2+54\lambda=0$

$\Rightarrow 6\lambda\left[\lambda^2-6\lambda+9\right]=0$

$\Rightarrow \lambda=0, \lambda=3$ [Distinct values]

So sum $=0+3=3$